COVID-19 Emo: Crowdsource Emotion Annotation for Anthropogenic Disasters

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***Abstract*—Since its discovery in late 2019, the COVID-19 pandemic has spread at an alarming speed, leaving millions of people infected. Countries worldwide have imposed lockdown to contain the disease, which leads to severe inhibition of in-person interactions. As a result, people have taken to social media (e.g. Twitter) to express their emotions and maintain daily relationships. These platforms provide researchers with the essential tools to study pandemic reactions over time. In this paper, we present our data processing techniques on the COVID-19 tweet dataset. Additionally, we introduce a Mechanical Turk crowdsource task that supports sentiment annotation on this dataset.**

I. Introduction

Anthropogenic disasters are the result of human negligence or failure of an artificial system. These disasters can range from violence (terrorism), industrial (nuclear explosion) to health hazards (COVID-19). Each type of disaster can bear a different impact on the ecosystem and human life. In particular, the COVID-19 pandemic, a health hazard, has been known to transform the way we go about our daily lives. Due to its nature as respiratory disease (airborne transmission), human-human interaction has helped spread the virus further and quicker. Therefore, governments around the world have taken measures to inhibit social interactions: quarantine, social distancing, mask enforcement. The only way to maintain relationships with other people is through social media (Twitter). These platforms have thus enabled researchers to understand the expression and perception of emotions at a significantly larger scale.

Emotion distributional shifts have reduced the effectiveness of current natural processing methods in the domain of disasters [1]. Detecting emotions in these events requires an adequate understanding of the implicit meanings, not just on the surface level. In this project, we aim to construct a semi-supervised method to capture the emotions of a given disaster (COVID-19 in particular).

We are working to introduce a dataset of COVID-19 related tweets (in English). 1,000 sampled tweets are labelled with eight emotions from the Plutchik Wheel of Emotions [2] (Figure 1). Amazon Mechanical Turk workers then perform annotations on this sampled dataset. Plutchik Emotion Agreement (PEA) score, assignment duration, and example response inspection are the metrics to understand inter-annotator agreement on these emotion labels.

II. Related Work

Our current work is largely based on previous work by Shrey et al. [1]. In this paper, they introduce HurricaneEmo, a hurricane-related tweet dataset, and the PEA metric. This study also proposes various pre-training forms to create strong biases and establishes domain adaptation baselines. In general, pre-trained models don't perform well on the Plutchik-8 emotion detection task, which they try to explain in an accompanying error analysis of the incorrectly predicted samples. In our project, we try to reproduce the methods in Shrey et al. study on the COVID-19 emotions.

III. Dataset Preprocessing

In this section, we present the COVID-19-TweetIDs dataset, which is the starting point for our dataset construction, (III.1) and discuss our data preprocessing techniques (III.2).

A. COVID-19-TweetIDs Dataset

Chen Emily et al. (2020) release an ongoing collection of tweets IDs associated with the COVID-19 pandemic, which commenced on January 28, 2020. The current version (2.9) contains about 450,000 tweets. For consistency, we refer to this dataset as COVID-19-TweetIDs.

B. Dataset Preprocessing

The COVID-19-TweetIDs dataset only publicizes tweet IDs to respect Twitter's term of service. Given this property, we use Twarc [3] to reconstruct the content of the tweets. We replace Twitter usernames and links with ⟨USER⟩ and ⟨URL⟩. We then utilize fastText [4] [5], a library for text classification and learning word representation, to determine the language of each tweet. In total, the dataset contains tweets in 48 different languages, with English being the most popular language (over 65% of the dataset).

We remove all non-words (emojis, symbols, etc.) using RegEx. We then remove all duplicate and NA values from the dataset. Finally, we delete all tweets that contain less than two words. To ensure that every tweet display at least one emotion type, we perform lexical filtering on the dataset using NRC Word-Emotion Association Lexicon (EmoLex) [6] [7]. EmoLex contains about 10,000 unigrams (words) in various languages with its emotion intensity score. We make sure that each tweet consists of at least one word derived from EmoLex. This method removes a significant amount of noise (articles

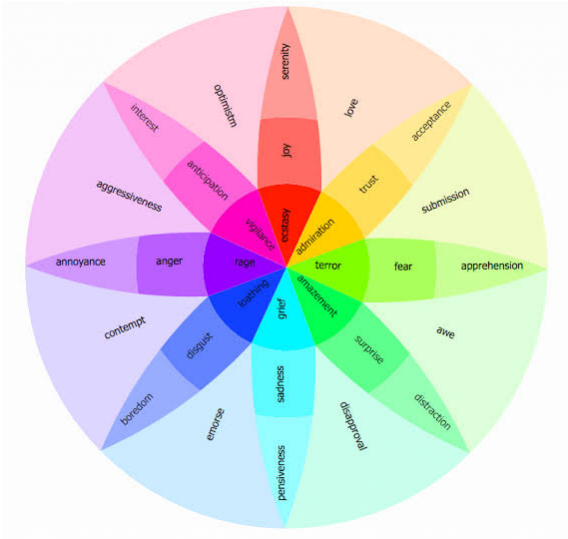

Fig. 1. Plutchik Emotion Wheel

that have no emotions) from the dataset, which makes dataset annotation easier. After performing these methods, the size of our dataset shrinks from about 32,000 down to 25,000 tweets.

IV. Crowdsource Annotation

In this section, we discuss the attributes of our Amazon Mechanical Turk task (IV.I), the process we follow to extract meaningful results from the responses (IV.II), and the conclusion after our pilot batch (IV.III).

A. Task Design

We randomly sample 1,000 tweets from the dataset for annotation. We design an annotation task with Boto3 and CrowdHTML to publish on Amazon Mechanical Turk. This task contains six tweets: five from the sample dataset, with an example at the end as a quality control. These examples and their sample annotations are in the task instruction, which filters out workers who annotate randomly or don't read the instructions. We request the workers to label the tweets with a list of Plutchik-8 emotions, including anger, anticipation, joy, trust, fear, surprise, sadness, disgust. If the tweet doesn't display any of the above sentiments, the worker should choose the 'None of the Above' (NA) option.

For further quality control, we require that workers reside in English-speaking countries (the United Kingdom, the United States of America, New Zealand, Australia), have completed 500+ HITs with an acceptance rate >95%. Each HIT is assigned to 5 different workers.

B. Result Processing

We remove all the annotated tweets that have no emotions from the final dataset. Tweets that display no emotion are those that receive more than 60% of "None of the Above" (NA) annotation. To prepare for the emotion-to-radian conversion, we delete individual NA annotations from the remaining tweets.

We use PEA metric [1] to calculate inter-annotator agreement. This method supports the scoring sets of categorical values. First, we superimpose a unit circle on the Plutchik wheel, representing each emotion with a polar coordinate. After converting the emotions into their respective radian values, we calculate the pairwise agreement between two annotators on the same tweet. Given worker 1 and 2 with the categorical annotation set of $\{e_x^{(i)}\}_{i=1}^{n}$ and $\{e_x^{(i)}\}_{j=1}^{m}$, the pairwise agreement $d(w_1, w_2)$ is calculated as:

$$\frac{1}{n}\sum_{i=1}^{n} max_j(\mid 1 - \frac{1}{\pi} \mid f(e_x^{(i)}) - f(e_y^{(j)}) \mid\mid)$$

where $\frac{1}{n}$ is a normalizing constant and f is a function to convert Plutchik emotions into their respective radian values. We use the agreement scores to further calculate per-worker PEA score, which is obtained by averaging over a worker's agreement score with the others. For example, provided that worker w_1 to w_3 annotate the same tweet, the PEA score for w_1 will be PEA$(w_1) = 1/2(d(w_1, w_2) + d(w_1, w_3))$. Workers whose PEA scores are in the bottom 5% range will have their submission manually inspected, which may lead to a rejection.

In addition to the PEA score, we use two other metrics to filter out sub-standard annotations. First, we examine workers' responses to the example tweets. By doing this, we ensure that workers follow the task instructions. Second, we look into the assignment's duration, or how long the workers spend on the task. We hypothesize that if a worker finishes the task in less than 5-10 seconds, there is a high chance that the submission will not meet the standard. We rank the workers according to their performance in these metrics, inspect their submission manually, and remove the under-qualified submission.

C. Pilot Batch

After testing the task on Amazon Mechanical Turk Sandbox, we release the pilot batch, which contains 100 tweets. This batch is finished in 20 minutes by 50 workers. The 95th percentile of the PEA score is about 0.44, which is 0.1 lower than the threshold in Shrey et al. paper [1]. Based on this result, we decide to improve the performance by adding two metrics: assignment duration and responses to example tweets.

V. Future Work

This ongoing project is currently in the second phase (out of four phases). The next step for this project will be the quantitative analysis of the annotated dataset. After finalizing the dataset, we will train a model to predict the sentiments of a given anthropogenic disaster.

VI. Acknowledgments

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